

BIOLOGY 101 · RECITATION HOMEWORK

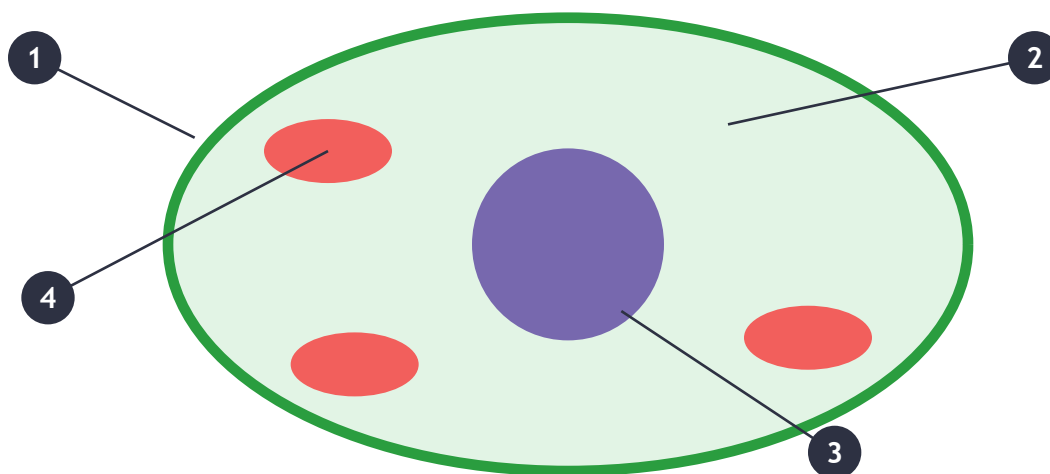
Homework 1: Cells

Name: _____

Date: _____

Instructions: About 20 minutes. Part A uses the diagram – the numbers point to parts of the cell. Look back at your notes if you need to. Total: **26 points**.

Part A — Label the cell



1. For each number on the diagram, write the part's name and its job (in your own words – the 'city' comparison is fine). (8 pts)

#	Part name	Its job
1		
2		
3		
4		

Part B — The ladder

2. Put these in order from SMALLEST to BIGGEST: organ, atom, cell, molecule, tissue, organism. (3 pts)

Part C — Doubling

3. One cell divides, and every new cell keeps dividing. Fill in how many cells there are after each round. (3 pts)

Rounds of division	0	1	2	3	4	5	6
Number of cells	1						

4. After 10 rounds there are 1,024 cells. About how many after 11 rounds? (1 pt)

Part D — Match the cell to its fact

5. Draw a line from each cell type to the fact that describes it. (4 pts)

Red blood cell	Stacks in layers to build a wall – only a few layers thick in a preemie
Skin cell	A single-celled living thing; some kinds cause infections like sepsis
Bacterium	Carries electrical signals around the body
Nerve cell	Throws out its nucleus to make room for hemoglobin; lives about 120 days

Part E — Power plants

6. When mitochondria burn sugar with oxygen, what is the “exhaust”? (1 pt)

- ☐ A. Carbon dioxide and water
- ☐ B. Nitrogen and oxygen
- ☐ C. Hemoglobin and bilirubin
- ☐ D. Salt and more sugar

Part F — Flashback to NICU Tutor

7. Why do newborns turn yellow (jaundice)? Use the words “red blood cells” and “bilirubin” in your answer. (2 pts)

8. The isolette keeps the air very humid for the tiniest babies. What does that protect against? (1 pt)

Part G — Think like a doctor

9. A preemie in the NICU is gaining about 20 grams every day. Using what you know about cells, explain why this baby needs so much sugar and oxygen. (3 pts)

Score: _____ / 26 points

Professor's comment:
